Questions

Q1: **Maps** are not shown.

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Q5: Countries on a choropleth map without values are plain white. This is bad for coastal countries because they disappear completely. How do I **indicate missing countries on a choropleth map with a special colour**? Furthermore, is it possible to change the **background** (ocean) colour?

Q6: Do we need to keep the **Actions button** in our Vega-Lite visualisation? If not, how can we remove it?

Q7: Why does my code work fine in **Vega Editor** but shows errors in **Visual Studio Code**?

Q8: After adding a colour scheme, the **bar chart sorting** is broken.

Q9: The Vega-Lite examples change the **colour of text annotations and labels**. How can I do this?.

Q10: For **Concat** charts, when I set the width to "**container**", the chart goes outside of my div.

Q11: When I have the whole earth map in my topojson file, do I use the whole file in Vega-Lite even if I only want a map of Australia? Is there a way to **clip map geometry** in **Mapshaper** and then download only that part?

Q12: I see different displays in **Live Server and HTML on GitHub**.

Q13: Visualisations works fine in Vega Editor or Visual Studio Code but are **not shown on the HTML page**.

*Q14: How to add an **annotation** to a stacked bar chart?*

Q1: Maps are not shown.

1. check whether you are using the raw data instead of the GitHub data link (https://edstem.org/au/courses/6022/discussion/601807)
2. Remove all calculations, log scales, etc. Choose a basic colour scale first, then again add calculations, log scales, colour scales, etc. and test each time you add a feature.
3. Double-check the following attributes:
    - "feature"
    - "lookup"
    - "key"
    - everything that you need for the CSV file must be entered into "fields".

Q2: Incorrect colours on a choropleth map.

Few possible solutions:

1. Make sure your CSV data is clean. If there is an empty space in front or behind of the country names, you won't be able to match the country name in the CSV and the Topojson.
2. Change the country name mapping from "NAME" to "NAME_LONG", which might be the standard used in the CSV file.
3. Make sure the URL in your Vega-Lite code is the raw GitHub user content link instead of a local path unless you are using Vega Viewer.

Q3: Unable to read CSV files uploaded on Git.

Click on Raw, and then use that raw data. The URL should be something like

https://raw.githubusercontent.com/username/FIT3179/main/XXX.csv

Q4: Topojson file from Mapshaper is very big

TopoJSON and GeoJSON are two special forms of JSON files. They are still in JSON format, so either .json or .topojson is fine.

The file extension is optional in Vega-Lite. By default, it uses the file extension to match the file type. But if you define the type as "topojson", then the file will be treated as topojson file no matter what extension (e.g., json, or geojson, or even .txt).

You can have a smaller file by simplifying the file with Mapshaper. To do so, click on "Simplify" on the top-right corner. Then select apply.

After that, drag the slider to simplify the file (e.g., to 10%).

If Mapshaper indicates an error with the data, click "Repair" on the top-left corner to fix the error.

Q5: Countries on a choropleth map without values are plain white. This is bad for coastal countries because they disappear completely. How do I indicate missing countries on a choropleth map with a special colour? Furthermore, is it possible to change the background (ocean) colour?

You can add another base map layer and show all empty countries with a grey colour.

The code is available at:
https://github.com/KaneSec/vega_lite/blob/main/3_choropleth_map/js/empty_data_map.vg.json

Also, for the background colour of the visualisation, you will need to add a config for the whole visualisation. Check the example here:

*https://kanesec.github.io/vega_lite/2021/1.%20Intro%20to%20vega-*lite/2.%20dupliacte%20the%20european%20city%20example/

You can add this before the last "}".

```
"config": {
"background": "#F3FAF9"
}
```

If you only want to highlight the ocean, you will need a topojson file that contains the ocean outlines.

Q6: Do we need to keep the Actions button in our Vega-Lite visualisation? If not, how can we remove it?

There is no need to include the Actions button. To remove it:

vegaEmbed('#vis', spec, {"actions": false})

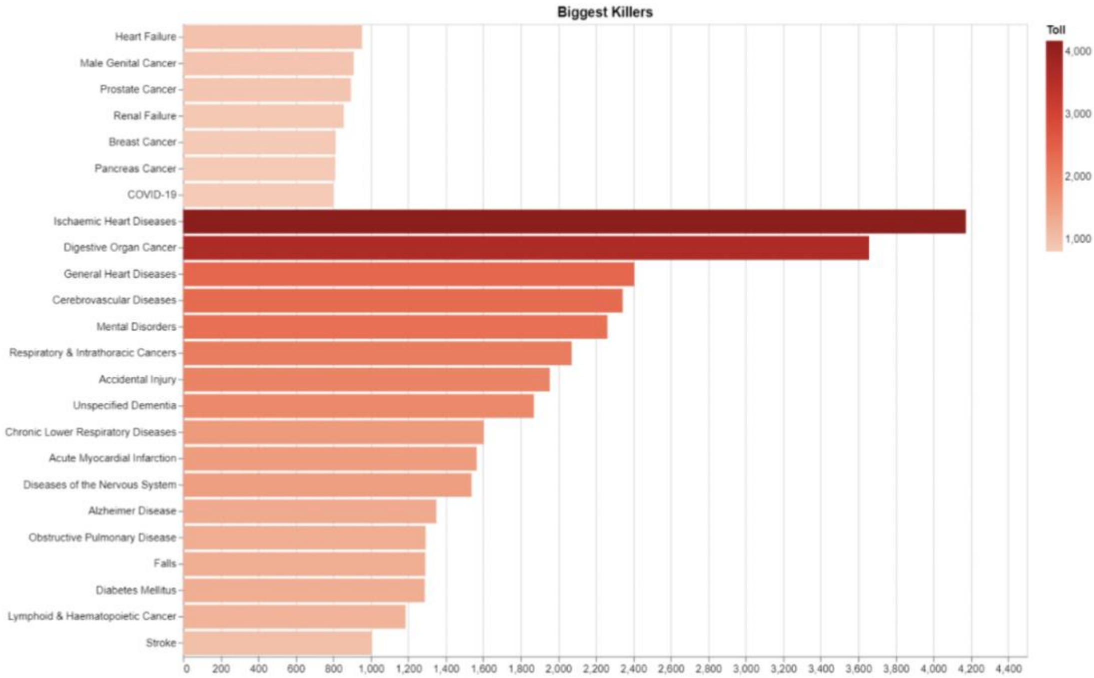
when you call the vegaEmbed function in your index.html file.

Q7: Why does my code work fine in Vega Editor but shows errors in Visual Studio Code?

The problem might be that the plugin in VS Code does not include the latest Vega-Lite compiler.

You might need to use Vega Editor for this part; or use "Open with Liver Server" in VS Code to check the result.

Q8: After adding a colour scheme, the bar chart sorting is broken




```
{
  "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
  "width": 800,
  "height": 600,
  "title": "Biggest Killers",
  "data": {"url": "https://raw.githubusercontent.com/fakeURL"},
  "mark": "bar",
  "encoding": {
      "x": {"field": "Toll", "type": "quantitative", "title": ""},
      "y": {"field": "Cause Of Death", "type": "nominal", "title":
"", "sort": "-x"},
      "color": {
        "field": "Toll",
        "type": "quantitative",
        "scale": {"scheme": "reds"}
    }
  }
}
```

After you add "color", Vega-Lite considers "Toll" as the original type "String".

Adding a transformation to change "Toll" to "int" at the start will solve the issue.

```
"transform": [
{"calculate":"parseInt(datum.Toll)", "as": "Toll"}
],
```

Q9: The Vega-Lite examples change the colour of text annotations and labels. How can I do this?

This is not possible with Vega-Lite, but possible with CSS.

Q10: For Concat charts, when I set the width to "container", the chart goes outside of my div.

Unfortunately, it will only align your main graph, not the legend.

There is some advanced JavaScript code to fix this. Check this earthquake data example if you are interested: https://github.com/KaneSec/earthquake

Q11: When I have the whole earth map in my topojson file, do I use the whole file in Vega-Lite even if I only want a map of Australia? Is there a way to clip map geometry in Mapshaper and then download only that part?

Few options here:

1. In Mapshaper, click on "console", then type the following command to clip the map geometry:
   ```
   -clip bbox=140,-50,146,-20
   ```
   Adjust latitude and longitude values to the boundary of your map.
2. In Vega-Lite, use
   ```
   "projection": {
       "type": "equirectangular",
       "center": [146, -36],
       "scale": 500
   }
   ```
   You can adjust the map type, center and scale. This defines the extent of your map, but does not clip the map geometry (so your web page may still download a large amount of unnecessary geometry data).
3. You can find an Australia map instead of a world map from ABS:

   https://www.abs.gov.au/AUSSTATS/abs@.nsf/DetailsPage/1270.0.55.001July

   %202016?OpenDocument

Q12: I see different displays in Live Server and HTML on GitHub.

Clear your browser cache, including cookies, and try again.

Q13: Visualisations works fine in Vega Editor or Visual Studio Code but are not shown on the HTML page.

Make sure you use the latest Vega JavaScripts:

```
<script src="https://cdn.jsdelivr.net/npm/vega@5.22.1"></script>
<script src="https://cdn.jsdelivr.net/npm/vega-lite@5.4.0"></script>
<script src="https://cdn.jsdelivr.net/npm/vega-embed@6.21.0"></script>
```

Q14: How to add an annotation to a stacked bar chart?

One solution is by defining a fake opacity encoding and then sort the data based on that. You need "value" for the text encoding, but the value is also reversed.

Check and run the code below:

```
{
    "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
    "width": 657, "height": 400,
    "background": "#292D36",
    "config": {
      "style": {
        "cell": {
          "stroke": "transparent"
        }
      },
      "axis": {"gridColor": "#414146", "titleColor":"#BEBECC",
"labelColor":"#BEBECC", "labelFontSize": 13 , "titleFontSize": 15},
      "legend": {"titleColor":"#BEBECC", "labelColor":"#BEBECC",
"orient": "top", "labelFontSize": 13 , "title": null}
    },
    "data": {"url":
"https://raw.githubusercontent.com/user0028/visualization2/main/Data/
owid_cleaned_stack.csv"},
    "transform": [
        {
            "filter": {"not": {"field": "location", "oneOf":
["World", "Asia", "Europe", "North America", "South America",
"Africa", "Oceania", "European Union", "Gibraltar", "Pitcairn"]}}
        },
        {
          "calculate":
"toNumber(datum.people_vaccinated_per_hundred)",
          "as": "people_vaccinated_per_hundred"
        },
        {
            "calculate":
"toNumber(datum.people_fully_vaccinated_per_hundred)",
            "as": "people_fully_vaccinated_per_hundred"
        },
        {
            "aggregate": [
                {"op": "max", "field":
"people_vaccinated_per_hundred", "as": "Total Vaccinated"},
```

```
                {"op": "max", "field":
"people_fully_vaccinated_per_hundred", "as": "Fully Vaccinated"}
              ],
            "groupby": ["location"]
        },
        {"calculate": "datum['Total Vaccinated']-datum['Fully
Vaccinated']", "as": "Partially Vaccinated"},
        {
            "window": [{
              "op": "rank",
              "as": "rank"
            }],
            "sort": [{ "field": "Total Vaccinated", "order":
"descending" }]
        },
        {
            "filter": "datum.rank <= 7"
        },
        {
            "fold": ["Partially Vaccinated", "Fully Vaccinated"]
        }
    ],
    "encoding": {
      "x": {
            "field": "value", "type": "quantitative",
"title":"Percentage Vaccinated",
            "axis":{"grid": false, "domain": false}
      },
      "y": {
                "field": "location", "type": "ordinal", "sort":
"opacity", "title":"Country"
            },
     "opacity": { "field": "rank", "scale":{
         "range":[1,1]
     }},
      "tooltip":[
        {"field": "Fully Vaccinated", "title": "% of People Fully
Vaccinated"},
        {"field": "Partially Vaccinated", "title": "% People of
Partially Vaccinated"},
        {"field": "Total Vaccinated", "title": "% Of People With At
Least One Dose"}
      ]
    },
    "layer": [
      {
        "mark": {"type": "bar"},
        "encoding": {
            "color":{
                    "field":"key",
                    "scale": {"scheme": "darkblue"}
                }
        }
      },{
      "transform": [
        {"calculate": "datum['Total Vaccinated'] - datum.value",
"as": "value_2"}
      ],
```

```
      "mark": {
          "type": "text",
          "align": "left",
          "baseline": "middle",
          "dx": 3,
          "color": "white"
      },
      "encoding": {
          "text": { "field": "value_2", "type": "quantitative" }
      }
  }]
}
```